

Overview

- Manage and analyze knitting related products.
 - Yarn
 - Knitting needles
 - Knitting accessories
 - Brand and supplier information
 - Inventory and product availability
- Main tasks of the database
 - Store product information
 - Track inventory levels
 - Identify the most cost effective knitting products
 - Allow updates to product yardage, weight, needle size, etc

Business Rules

- Yardage for yarn must be greater than zero
- Product availability must be greater than zero
- If a supplier does not have the item in stock it must be removed
- Full description of an item must be provided
- Needles must have a size, material, brand, and length

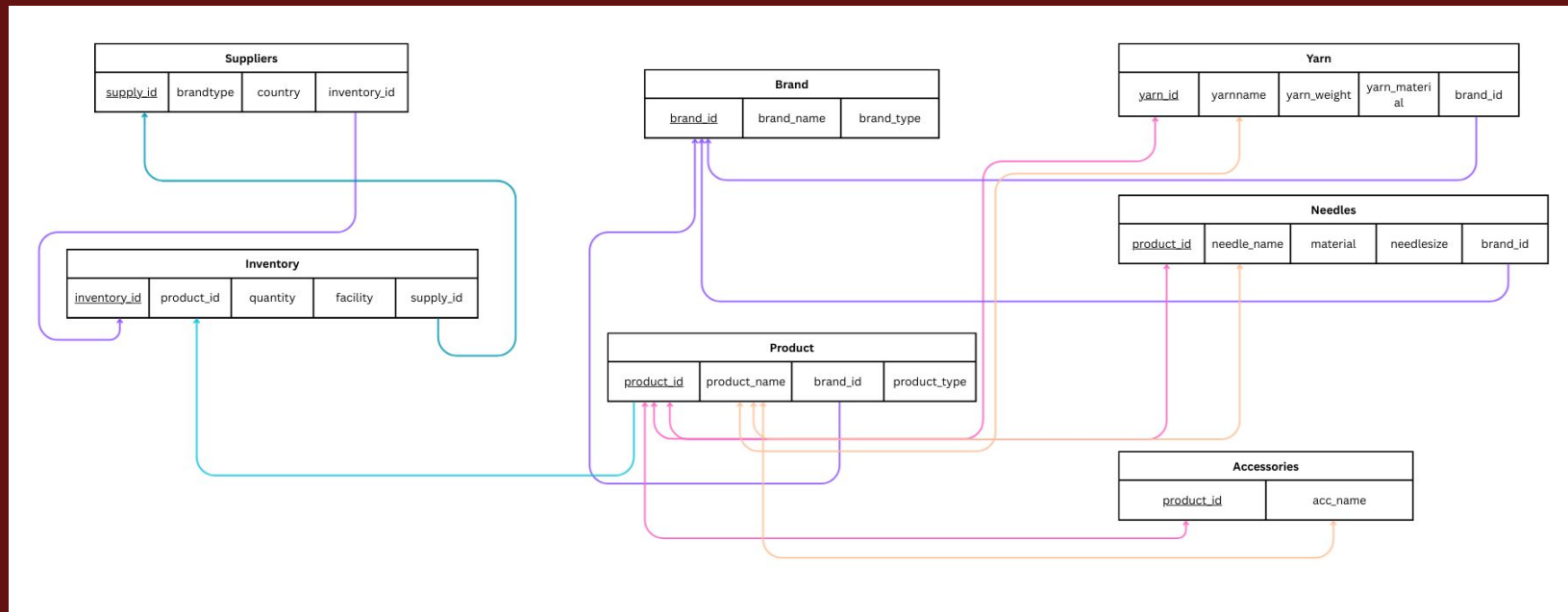
Entities

- Product (Supertype): Represents all knitting items
- Yarn (Subtype): The different weights of yarn available.
- Needles (Subtype): The different types and sizes of needles available.
- Accessories (Subtype): Additional knitting tools and items.
- Brand: Companies that produce the products
- Suppliers: Vendors that supply the products.
- Inventory: Storage information and stock levels.

Relationships

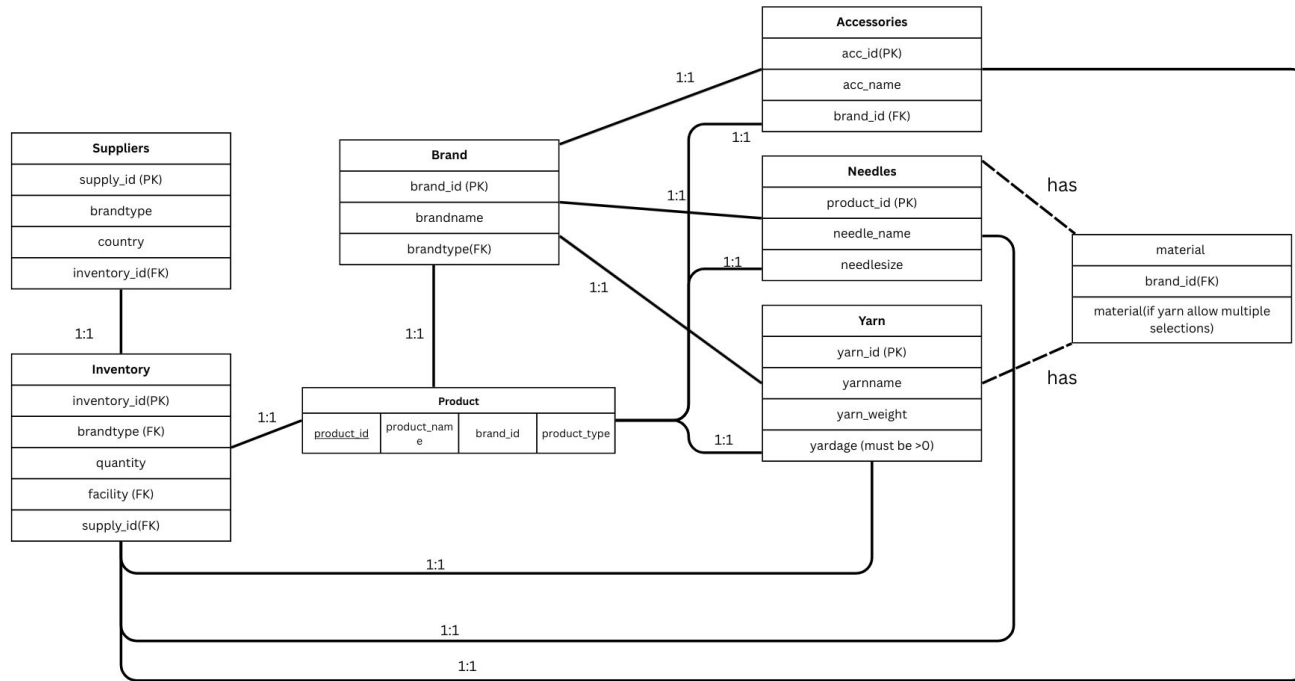
- Brand > Product: One brand can have many products, each product must belong to one brand
- Product > Yarn/Needles/Accessories: Product is a supertype, and each product must belong to exactly one subtype
- Suppliers > Inventory: A supplier may provide inventory for many products.

Relational Schemas



ER Diagram

6



Populated Tables

brand_id (pk)	brand_name	brand_type
1	chaigoo	needles
2	lykke	needles
3	clover	needles
4	addi	needles
5	knitters pride	needles
6	red heart	yarn
7	lion brand	yarn
8	loops and threads	yarn
9	berroco	yarn
10	cascade	yarn
11	rowan	yarn
12	malabrigo	yarn
13	urth	yarn
14	ito	yarn
15	noro	yarn

product_id	acc_name
31	ball winder
32	swift
33	scissors
34	blue button
35	yellow button
36	cable needle
37	darning needle
38	stitch marker
39	needle protector
40	yarn bag

Populated Tables 2

product_id	needle_name	material	needlesize	brand_id
16	chaigoo	wood	1.5	1
17	lykke	bamboo	1.75	2
18	clover	metal	2	3
19	addi	plastic	2.25	4
20	knitters pride	wood	2.5	5
21	chaigoo	bamboo	2.75	1
22	lykke	metal	3	2
23	clover	plastic	3.25	3
24	addi	wood	3.5	4
25	knitters pride	bamboo	3.75	5
26	chaigoo	metal	4	1
27	lykke	plastic	4.5	2
28	clover	wood	5	3
29	addi	bamboo	5.5	4
30	knitters pride	metal	6	5

Populated Tables 3

yarn_id	yarnname	yarn_weight	yardage	yarn_material	brand_id
1	super soft	aran	220	acrylic, polyester	7
2	ariel	lace	90	mohair	9
3	cotton wow	dk	115	cotton	6
4	silk dream	fingering	400	silk	14
5	impeccable yarn	worsted	285	acrylic	8
6	amble	worsted	280	wool, alpaca	10
7	baby blossom	bulky	170	acrylic, nylon	11
8	balayage	sport	190	baby alpaca, virgin wool	12
9	black and white	fingering	400	superwash merino, nylon	9
10	chenillovely	dk	142	polyester	6
11	costal	worsted	185	cotton	8
12	rios	aran	210	superwash merino wool	12
13	natsumeki	super bulky	525	cotton, silk, polyamide	15
14	silky wool	sport	192	wool, silk, nylon	13
15	soft donegal	bulky	210	merino	11

Populated Tables 4

product_id	product_name	brand_id	product_type
1	super soft	7	yarn
2	ariel	9	yarn
3	cotton wow	6	yarn
4	silk dream	14	yarn
5	impeccable yarn	8	yarn
6	amble	10	yarn
7	baby blossom	11	yarn
8	balayage	12	yarn
9	black and white	9	yarn
10	chenillovely	6	yarn
11	costal	8	yarn
12	rios	12	yarn
13	natsumeki	15	yarn
14	silky wool	13	yarn
15	soft donegal	11	yarn

16	chaigoo	1	needles
17	lykke	2	needles
18	clover	3	needles
19	addi	4	needles
20	knitters pride	5	needles
21	chaigoo	1	needles
22	lykke	2	needles
23	clover	3	needles
24	addi	4	needles
25	knitters pride	5	needles
26	chaigoo	1	needles
27	lykke	2	needles
28	clover	3	needles
29	addi	4	needles
30	knitters pride	5	needles

Populated Tables 5

31	ball winder	1	accessories
32	swift	2	accessories
33	scissors	3	accessories
34	blue button	4	accessories
35	yellow button	5	accessories
36	cable needle	6	accessories
37	darning needle	7	accessories
38	stitch marker	8	accessories
39	needle protector	9	accessories
40	yarn bag	10	accessories

Populated Tables 6

supply_id	brandtype	country
123	accessories	china
456	needles	taiwan
789	yarn	america
101	yarn	ireland

product_id	quantity	facility	supply_id
31	46	shanghai	123
32	27	shanghai	123
33	16	suzhou	123
34	18	tianjin	123
20	11	taipei	456
21	30	taipei	456
22	52	taipei	456
1	33	chicago	789
2	19	chicago	789
3	29	dallas fort worth	789
4	34	dallas fort worth	789
5	44	galway	101
6	56	dublin	101
7	71	galway	101

Queries

13

All products with their brand names

```
SELECT p.product_id, p.product_name, b.brand_name, p.product_type FROM Product p JOIN Brand b ON p.brand_id = b.brand_id;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> | ☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

product_id	product_name	brand_name	product_type
16	chaigoo	chaigoo	needles
21	chaigoo	chaigoo	needles
26	chaigoo	chaigoo	needles
31	ball winder	chaigoo	accessory
36	cable needle	chaigoo	accessory
17	lykke	lykke	needles
22	lykke	lykke	needles
27	lykke	lykke	needles
32	swift	lykke	accessory
37	darning needle	lykke	accessory
18	clover	clover	needles
23	clover	clover	needles
28	clover	clover	needles
33	scissors	clover	accessory
38	stitch marker	clover	accessory
19	addi	addi	needles
24	addi	addi	needles
29	addi	addi	needles
34	blue button	addi	accessory
39	needle protector	addi	accessory
20	knitters pride	knitters pride	needles
25	knitters pride	knitters pride	needles

yarn products with weight and yardage

Showing rows 0 - 14 (15 total, Query took 0.0007 seconds.)

```
SELECT p.product_name, y.yarn_weight, y.yardage, y.yarn_material FROM yarn y JOIN product p ON y.yarn_id = p.product_id;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25

Extra options

product_name	yarn_weight	yardage	yarn_material
super soft	aran	220	acrylic, polyester
ariel	lace	90	monhair
cotton wow	dk	115	cotton
silk dream	fingering	400	silk
impeccable yarn	worsted	285	acrylic
amble	worsted	280	wool, alpaca
baby blossom	bulky	170	acrylic, nylon
balayage	sport	190	baby aplaca, virgin wool
black and white	fingering	400	superwash merino, nylon
chenilovely	dk	142	polyester
costal	worsted	185	cotton
rios	aran	210	superwash marino wool
natsumeki	super bulky	525	cotton, silk, polyamide
silk wool	sport	192	wool, silk, nylon
soft donegal	bulky	210	merino

Queries (cont'd)

14

inventory quantity for products

```
SELECT p.product_name, i.quantity, i.facility FROM Inventory i JOIN Product p ON  
i.product_id = p.product_id;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 ▾

Extra options

product_name	quantity	facility
ball winder	46	shanghai
swift	27	shanghai
scissors	16	suzhou
blue button	18	tianjin
knitters pride	11	taipei
chaigoo	30	taipei
lykke	52	taipei
super soft	33	chicago
ariel	19	chicago
cotton wow	29	dallas fort worth
silk dream	34	dallas fort worth
impeccable yarn	44	galway
amble	56	dublin
baby blossom	71	galway

Knitting needles and their material and size

```
SELECT n.needle_name, n.material, n.needlesize FROM needles n;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[C \]](#)

☐ Show all | Number of rows: 25 ▾

Extra options

needle_name	material	needlesize
chaigoo	wood	1.5
lykke	bamboo	1.75
clover	metal	2
addi	plastic	2.25
knitters pride	wood	2.5
chaigoo	bamboo	2.75
lykke	metal	3
clover	plastic	3.25
addi	wood	3.5
knitters pride	bamboo	3.75
chaigoo	metal	4
lykke	plastic	4.5
clover	wood	5
addi	bamboo	5.5
knitters pride	metal	6

Queries (cont'd)

Showing rows 0 - 9 (10 total, Query took 0.0004 seconds.)

```
SELECT a.acc_name, b.brand_name, b.brand_type FROM accessories a JOIN product p ON  
a.product_id = p.product_id JOIN brand b ON p.brand_id = b.brand_id;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 ▾

Extra options

acc_name	brand_name	brand_type
ball winder	chaigoo	needles
swift	lykke	needles
scissors	clover	needles
blue button	addi	needles
yellow button	knitters pride	needles
cable needle	chaigoo	needles
darning needle	lykke	needles
stitch marker	clover	needles
needle protector	addi	needles
yarn bag	knitters pride	needles

Showing rows 0 - 5 (6 total, Query took 0.0003 seconds.)

```
SELECT p.product_name, i.quantity FROM inventory i JOIN product p ON i.product_id  
= p.product_id WHERE i.quantity < 30;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 ▾

Extra options

product_name	quantity
swift	27
scissors	16
blue button	18
knitters pride	11
ariel	19
cotton wow	29

Queries (cont'd)

```
SELECT yarnname, yardage, yarn_weight FROM yarn WHERE yardage > 250;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP](#)]

☐ Show all | Number of rows: 25 ▾

Extra options

yarnname	yardage	yarn_weight
silk dream	400	fingering
impeccable yarn	285	worsted
amble	280	worsted
black and white	400	fingering
natsumeki	525	super bulky

```
SELECT p.product_name, s.supply_id, s.country FROM product p JOIN inventory i ON  
p.product_id = i.product_id JOIN suppliers s ON i.supply_id = s.supply_id;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 ▾

Extra options

product_name	supply_id	country
impeccable yarn	101	ireland
amble	101	ireland
baby blossom	101	ireland
ball winder	123	china
swift	123	china
scissors	123	china
blue button	123	china
knitters pride	456	taiwan
chaigoo	456	taiwan
lykke	456	taiwan
super soft	789	america
ariel	789	america
cotton wow	789	america
silk dream	789	america

Queries (cont'd)

```
SELECT needle_name, material, needlesize FROM Needles WHERE material = 'Bamboo';
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refr](#)

☐ Show all | Number of rows: 25 ▼

Extra options

needle_name	material	needlesize
lykke	bamboo	1.75
chaigoo	bamboo	2.75
knitters pride	bamboo	3.75
addi	bamboo	5.5

Conclusion

Difficulties

- An online knitting store has many moving parts
- If one aspect of an entity is missing it messes up the rest of the supply chain
- Most data depends on each other and needs a source to rely on
- There were a lot of different entities to fill

Room for improvement

- A better understanding of the entities
- How to effectively create data